

Semantics of Comparative Constructions

Class 3: Comparatives, finally

Filipe Hisao Kobayashi

University of Salzburg

3 August 2026

OVA Summer School · University of Debrecen

Part 1

Recap & today's plans

Two important that will come up today

1) Adjectives have relate individuals to degrees via measure functions and the $\geq$ -relation:

$$(1) \quad \llbracket \text{tall} \rrbracket = \lambda d. \lambda x. \mathbf{tall}(x) \geq d$$

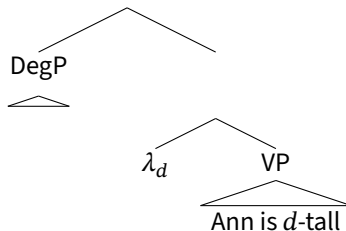
Two important that will come up today

1) Adjectives have relate individuals to degrees via measure functions and the $\geq$ -relation:

$$(1) \quad \llbracket \text{tall} \rrbracket = \lambda d. \lambda x. \mathbf{tall}(x) \geq d$$

2) By moving DegPs we can create **interval** denoting constituents:

(2) Ann is [DegP tall] $\Rightarrow$



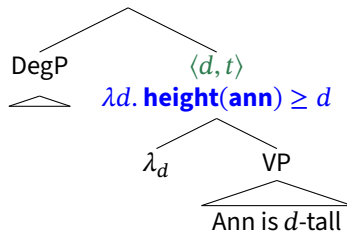
Two important that will come up today

1) Adjectives have relate individuals to degrees via measure functions and the $\geq$ -relation:

$$(1) \llbracket \text{tall} \rrbracket = \lambda d. \lambda x. \mathbf{tall}(x) \geq d$$

2) By moving DegPs we can create **interval** denoting constituents:

(2) Ann is [DegP tall] $\Rightarrow$



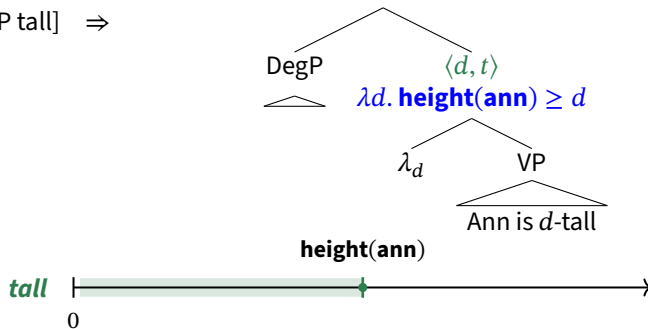
Two important that will come up today

1) Adjectives have relate individuals to degrees via measure functions and the $\geq$ -relation:

$$(1) \quad \llbracket \text{tall} \rrbracket = \lambda d. \lambda x. \mathbf{tall}(x) \geq d$$

2) By moving DegPs we can create **interval** denoting constituents:

(2) Ann is [DegP tall] $\Rightarrow$



Recap of antonyms ant their intervals

- Our meanings for antonym pairs:

(3) a. $\llbracket \text{tall} \rrbracket = \lambda d. \lambda x. \mathbf{height}(x) \geq d$
 b. $\llbracket \text{short} \rrbracket = \lambda d. \lambda x. d \geq \mathbf{height}(x)$

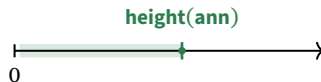
Recap of antonyms and their intervals

- Our meanings for antonym pairs:

- (3) a. $\llbracket \text{tall} \rrbracket = \lambda d. \lambda x. \mathbf{height}(x) \geq d$
b. $\llbracket \text{short} \rrbracket = \lambda d. \lambda x. d \geq \mathbf{height}(x)$

- Moving DegP from Spec,AP creates different intervals:

- (4) a. DegP λ_d [Ann is t_d -tall]



- b. DegP λ_d [Ann is t_d -short]



Today

- We'll finally discuss comparatives:
 - * the syntax of clausal comparatives
 - * semantics of *er*
 - * cross-polar anomalies & more

Part 2

The syntax of comparatives I: on *more & than*

The parts of a English comparative

- The three components of an English comparative:

Ducks are	more	friendly	[than geese].
	COMPARATIVE MARKER	<i>scalar predicate</i>	STANDARD PHRASE

The parts of a English comparative

- The three components of an English comparative:

Ducks are **more** **friendly** [**than** geese].
 COMPARATIVE MARKER *scalar predicate* STANDARD PHRASE

- **First challenge:** what's the dependency between *more* and the *than*-phrase?

The parts of a English comparative

- The three components of an English comparative:

Ducks are **more** **friendly** [**than** geese].
COMPARATIVE MARKER *scalar predicate* STANDARD PHRASE

- **First challenge:** what's the dependency between *more* and the *than*-phrase?
 - They can be very far from each other:
- (5) a. Ann was **more** quiet [**than** Beth].
b. Ann was **more** quiet when I met her [**than** Beth].
c. Ann was **more** quiet in the mornings when I met her [**than** Beth].

The parts of a English comparative

- The three components of an English comparative:

Ducks are **more** **friendly** [**than** geese].
COMPARATIVE MARKER *scalar predicate* STANDARD PHRASE

- **First challenge:** what's the dependency between *more* and the *than*-phrase?
 - They can be very far from each other:
- (5) a. Ann was **more** quiet [**than** Beth].
b. Ann was **more** quiet when I met her [**than** Beth].
c. Ann was **more** quiet in the mornings when I met her [**than** Beth].
- And *than*-phrases can't appear on their own:
- (6) *It rained [**than** yesterday].

Selectional restrictions

- Comparative markers seems to (L-)select standard markers:
(7) a. Ann is nic-**er** to me **than**/***as**/***that** Beth.
b. Ann knows **more** show tunes **than**/***as**/***that** Beth.
c. Cleo is **less** sporty **than**/***as**/***that** Deb.
d. Ella is **as** blunt **as**/***than**/***that** Flo.
e. Gabby made **fewer** cocktails **than**/***as**/***that** Holly.
- This is creates difficulties, since selection is often thought to be local.

Reducing number of selectional restrictions

- Consider the following paradigm from Selkirk 1970:

(8)	as	much	bread		as	little	bread
	too	much	bread		too	little	bread
	that	much	bread		that	little	bread
	so	much	bread		so	little	bread
	er	much	bread [> more]		er	little	bread [> less]

- Building on such data, Bresnan 1973 (and others before her) proposes the following:

	<i>er + many/much</i>	=	<i>more</i>
(9)	<i>er + little</i>	=	<i>less</i>
	<i>er + few</i>	=	<i>fewer</i>

Reducing number of selectional restrictions

- Consider the following paradigm from Selkirk 1970:

(8)	as	much	bread		as	little	bread
	too	much	bread		too	little	bread
	that	much	bread		that	little	bread
	so	much	bread		so	little	bread
	er	much	bread [> more]		er	little	bread [> less]

- Building on such data, Bresnan 1973 (and others before her) proposes the following:

	<i>er + many/much</i>	=	<i>more</i>
(9)	<i>er + little</i>	=	<i>less</i>
	<i>er + few</i>	=	<i>fewer</i>

- We can thus reduce selectional restrictions to **two**:
 - er...than*
 - as...as*

than*-phrases as arguments of *-er

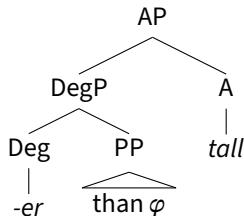
- We can account for these facts as follows:

(Chomsky 1965, Bresnan 1973, Heim 2001)

than-phrases as arguments of *-er*

- We can account for these facts as follows: (Chomsky 1965, Bresnan 1973, Heim 2001)
 - *than*-phrases are **arguments** of the comparative marker

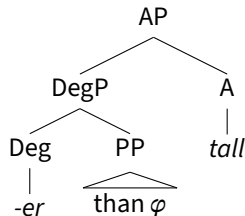
(10)



than-phrases as arguments of -er

- We can account for these facts as follows: (Chomsky 1965, Bresnan 1973, Heim 2001)
 - *than*-phrases are **arguments** of the comparative marker

(10)



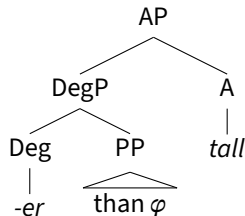
- but they must be obligatorily **extraposed**:

(11) I read [a book **that Ann gave me**] last year.

than-phrases as arguments of -er

- We can account for these facts as follows: (Chomsky 1965, Bresnan 1973, Heim 2001)
 - *than*-phrases are **arguments** of the comparative marker

(10)

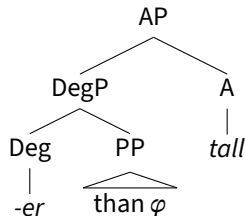


- but they must be obligatorily **extraposed**:
- (11) I read [a book **that Ann gave me**] last year.
→ I read [a book t_1] last year [**that Ann gave me**]₁

than-phrases as arguments of -er

- We can account for these facts as follows: (Chomsky 1965, Bresnan 1973, Heim 2001)
 - *than*-phrases are **arguments** of the comparative marker

(10)

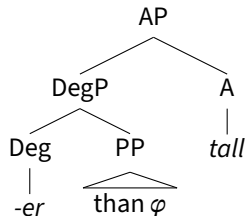


- but they must be obligatorily **extraposed**:
- (11) I read [a book **that Ann gave me**] last year.
→ I read [a book t_1] last year [**that Ann gave me**]₁
- (12) Ann is [[er **than Beth**] tall]

than-phrases as arguments of -er

- We can account for these facts as follows: (Chomsky 1965, Bresnan 1973, Heim 2001)
 - *than*-phrases are **arguments** of the comparative marker

(10)

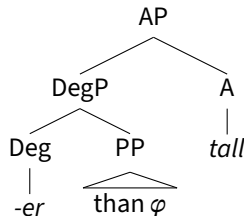


- but they must be obligatorily **extraposed**:
- (11) I read [a book **that Ann gave me**] last year.
→ I read [a book t_1] last year [**that Ann gave me**]₁
- (12) Ann is [[er **than Beth**] tall]
→ Ann is [[er t_1] tall] [**than Beth**]₁

than-phrases as arguments of -er

- We can account for these facts as follows: (Chomsky 1965, Bresnan 1973, Heim 2001)
 - *than*-phrases are **arguments** of the comparative marker

(10)



- but they must be obligatorily **extraposed**:
- (11) I read [a book **that Ann gave me**] last year.
→ I read [a book t_1] last year [**that Ann gave me**]₁
- (12) Ann is [[er **than Beth**] tall]
→ Ann is [[er t_1] tall] [**than Beth**]₁

(see Bhatt and Pancheva 2004 a slightly different proposal that accounts for this)

Part 3

The syntax of comparatives II: The silent structure of *than*-clauses

Clausal comparatives

- In English, the complement of *than* are sometimes unambiguously clauses:

- (13) a. Ann is taller than Beth.
b. Ann is taller than Beth is.

Clausal comparatives

- In English, the complement of *than* are sometimes unambiguously clauses:

- (13) a. Ann is taller than Beth.
b. Ann is taller than Beth is.

- When they're clausal, they seem to involve ellipsis:

(see Bresnan 1973)

- (14) a. * Ann seems taller than Beth seems tall.
b. Ann seems taller than Beth does ~~seem tall~~.
c. Ann seems than Beth seems ~~tall~~

VP-ELLIPSIS

COMPARATIVE DELETION

Clausal comparatives

- In English, the complement of *than* are sometimes unambiguously clauses:

- (13) a. Ann is taller than Beth.
b. Ann is taller than Beth is.

- When they're clausal, they seem to involve ellipsis: (see Bresnan 1973)

- (14) a. * Ann seems taller than Beth seems tall.
b. Ann seems taller than Beth does ~~seem tall~~. VP-ELLIPSIS
c. Ann seems than Beth seems ~~tall~~ COMPARATIVE DELETION

- There are cases that don't seem to involve ellipsis:

- (15) The table is longer than the door is wide. COMPARATIVE SUB-DELETION

- However, Bresnan 1973 argues that these still involve ellipsis:

- (16) a. Mary is happy with her work, and John is with his children.
b. * Mary's happy with her work, and John's with his children.
- (17) * The table's longer than the door's wide.

Phrasal *than*-clauses as ellipsis

- When *than* takes a non-clausal XP, it can have different “correlates” in the main clause:

(18) Ann lived closer to a supermarket last year [than **Beth**]
[than to **the university**]
[than **this year**]

Phrasal *than*-clauses as ellipsis

- When *than* takes a non-clausal XP, it can have different “correlates” in the main clause:

(18) Ann lived closer to a supermarket last year [than **Beth**]
[than to **the university**]
[than **this year**]

- We could, however, reduce all these cases as clausal complements + ellipsis:

(19) Ann lived closer to a supermarket last year [than **Beth** lived close to a supermarket last year]
[than Beth lived close to **the university** last year]
[than Beth lived close to a supermarket **this year**]

Phrasal *than*-clauses as ellipsis

- When *than* takes a non-clausal XP, it can have different “correlates” in the main clause:

(18) Ann lived closer to a supermarket last year [than **Beth**]
[than to **the university**]
[than **this year**]

- We could, however, reduce all these cases as clausal complements + ellipsis:

(19) Ann lived closer to a supermarket last year [than **Beth** ~~lived close to a supermarket last year~~]
[than ~~Beth lived close to~~ **the university** ~~last year~~]
[than ~~Beth lived close to a supermarket~~ **this year**]

- At least for now, we will assume exactly this.

Dependencies within the *than*-clause

- The elided adjective in the *than* clause can be deeply embedded:

- (20)
- a. Ann is taller than [Beth is ~~tall~~].
 - b. Ann is taller than [Cleo said [Beth is ~~tall~~]].
 - c. Ann is taller than [Cleo is believed [to have said [Beth is ~~tall~~]]].

Dependencies within the *than*-clause

- The elided adjective in the *than* clause can be deeply embedded:

- (20)
- a. Ann is taller than [Beth is ~~tall~~].
 - b. Ann is taller than [Cleo said [Beth is ~~tall~~]].
 - c. Ann is taller than [Cleo is believed [to have said [Beth is ~~tall~~]]].

- But it cannot be inside an island:

- (21) * Ann is taller than [I know [a student who is ~~tall~~]].
*see also: *What do you know a student who ate?*

Dependencies within the *than*-clause

- The elided adjective in the *than* clause can be deeply embedded:

- (20)
- a. Ann is taller than [Beth is ~~tall~~].
 - b. Ann is taller than [Cleo said [Beth is ~~tall~~]].
 - c. Ann is taller than [Cleo is believed [to have said [Beth is ~~tall~~]]].

- But it cannot be inside an island:

- (21) * Ann is taller than [I know [a student who is ~~tall~~]].
see also: **What do you know a student who ate?*

- What can we infer from this?

Dependencies within the *than*-clause

- The elided adjective in the *than* clause can be deeply embedded:

- (20)
- a. Ann is taller than [Beth is ~~tall~~].
 - b. Ann is taller than [Cleo said [Beth is ~~tall~~]].
 - c. Ann is taller than [Cleo is believed [to have said [Beth is ~~tall~~]]].

- But it cannot be inside an island:

- (21) * Ann is taller than [I know [a student who is ~~tall~~]].
*see also: *What do you know a student who ate?*

- What can we infer from this?
- **Chomsky 1977:** *than*-clauses involve (A-bar) movement of a null operator

Dependencies within the *than*-clause

- The elided adjective in the *than* clause can be deeply embedded:

- (20)
- a. Ann is taller than [Beth is ~~tall~~].
 - b. Ann is taller than [Cleo said [Beth is ~~tall~~]].
 - c. Ann is taller than [Cleo is believed [to have said [Beth is ~~tall~~]]].

- But it cannot be inside an island:

- (21) * Ann is taller than [I know [a student who is ~~tall~~]].
*see also: *What do you know a student who ate?*

- What can we infer from this?
- **Chomsky 1977:** *than*-clauses involve (A-bar) movement of a null operator
⇒ *But from where?*

Dependencies within the *than*-clause

- The elided adjective in the *than* clause can be deeply embedded:

- (20) a. Ann is taller than [Beth is ~~tall~~].
b. Ann is taller than [Cleo said [Beth is ~~tall~~]].
c. Ann is taller than [Cleo is believed [to have said [Beth is ~~tall~~]]].

- But it cannot be inside an island:

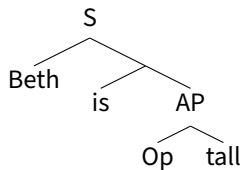
- (21) * Ann is taller than [I know [a student who is ~~tall~~]].
*see also: *What do you know a student who ate?*

- What can we infer from this?
- **Chomsky 1977**: *than*-clauses involve (A-bar) movement of a null operator
⇒ *But from where?*
- Sub-comparatives, again:

- (22) a. The window is wider than it is tall.
b. The window is wider than I thought it is tall.
c. * The window is wider than I bought a door that was tall.

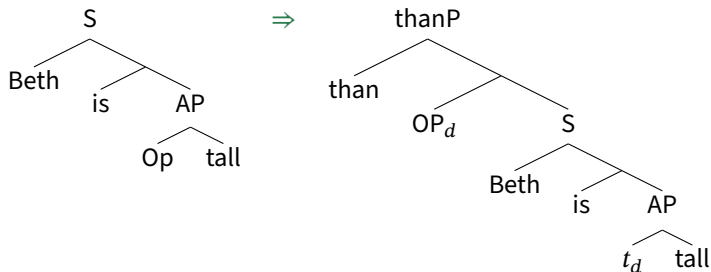
DegP movement, again

- We'll assume that clausal *than*-clauses involves the movement of a null operator in the same position as DegP:



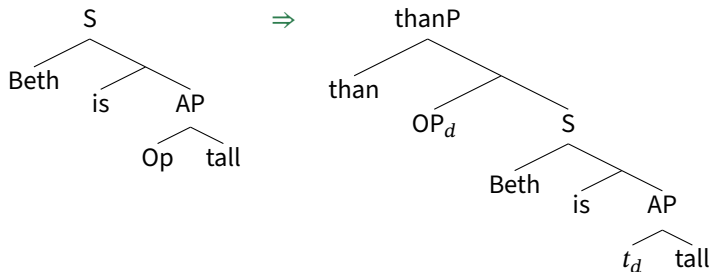
DegP movement, again

- We'll assume that clausal *than*-clauses involves the movement of a null operator in the same position as DegP:



DegP movement, again

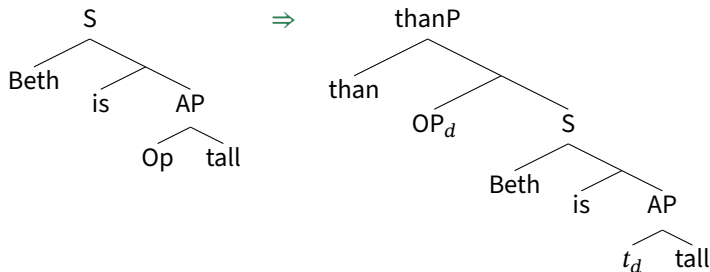
- We'll assume that clausal *than*-clauses involves the movement of a null operator in the same position as DegP:



- This allows for a uniform account of the meaning of the sister of *than*:

DegP movement, again

- We'll assume that clausal *than*-clauses involves the movement of a null operator in the same position as DegP:



- This allows for a uniform account of the meaning of the sister of *than*: **an interval** $\langle d, t \rangle$

(23) $\llbracket \text{thanP} \rrbracket = \lambda d. \mathbf{height}(\mathbf{beth}) \geq d$

Uniform analysis of clausal *than*P as intervals

- Again:

(24) Ann lived closer to a supermarket last year [than **Beth**]
[than to **the university**]
[than **this year**]

Uniform analysis of clausal *than*P as intervals

- Again:

(24) Ann lived closer to a supermarket last year [than **Beth**]
[than to **the university**]
[than **this year**]

- All these different *than*P denote intervals:

(25) a. $\text{than OP } [\lambda_d \text{ **Beth** lived d-close to a supermarket last year}]$
b. $\text{than OP } [\lambda_d \text{ Ann lived d-close to **the university** last year}]$
c. $\text{than OP } [\lambda_d \text{ Ann lived d-close to a supermarket **last year**}]$

Part 4

The syntax of comparatives III: Putting everything together

Recap I: the position of the *than*-clause

- We now derive the structure for the following sentence:

(26) Ann is taller than Beth.

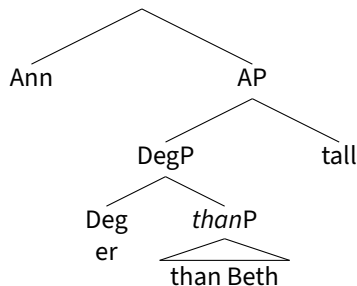
Recap I: the position of the *than*-clause

- We now derive the structure for the following sentence:

(26) Ann is taller than Beth.

- 1) The *than*-clause is generated as the complement of *er*:

(27)

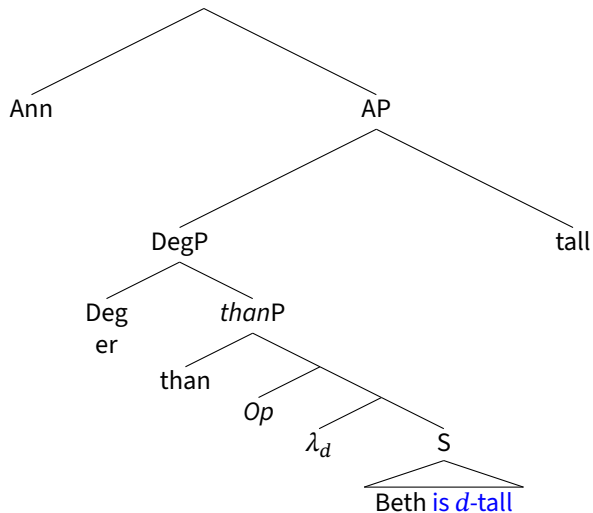


- We will ignore extraposition in our LFs.

Recap II: ellipsis in the *than*-clause

2) there's ellipsis & movement in the *than*-clause:

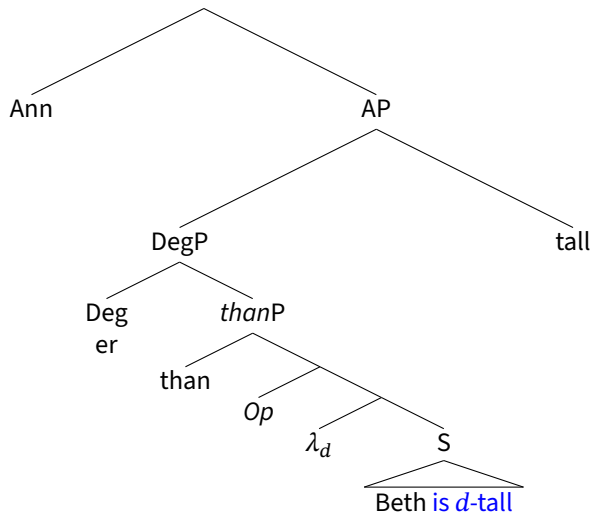
(28)



Recap II: ellipsis in the *than*-clause

2) there's ellipsis & movement in the *than*-clause:

(28)

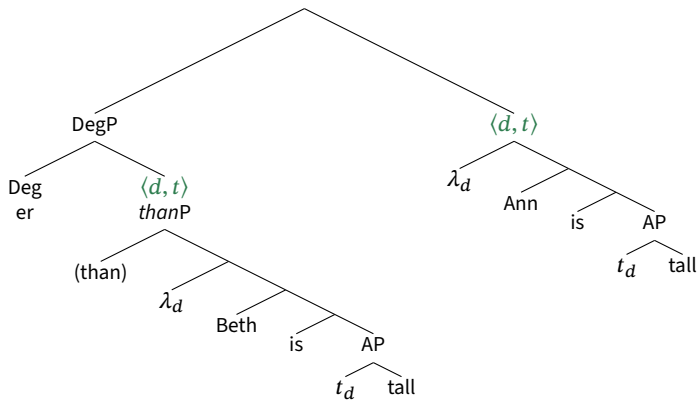


- But what's the antecedent of the ellipsis?

Final piece: movement of DegP

- **Proposal:** the DegP in the matrix clause also moves:

(29) Ann is [er [than Beth is tall]] tall →



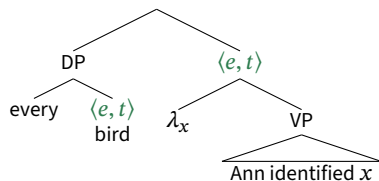
Part 5

The semantics of comparatives

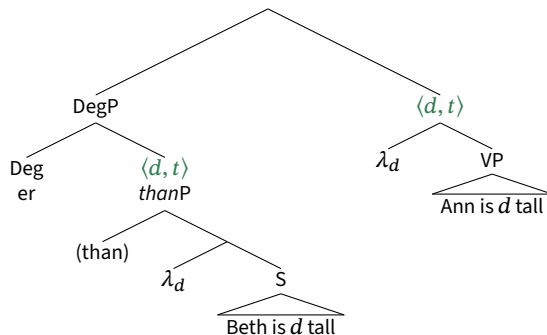
Determiners & *er*

- Compare the following two structures:

(30)



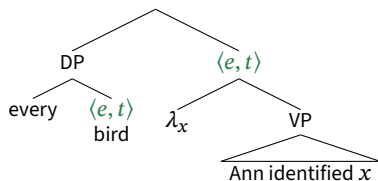
(31)



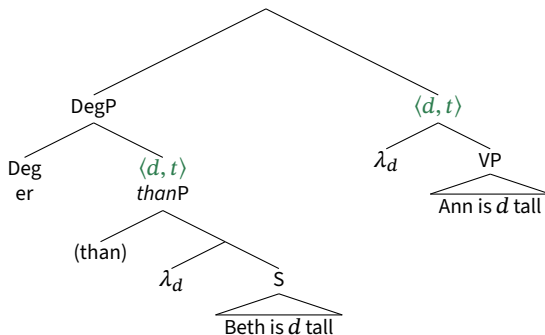
Determiners & *er*

- Compare the following two structures:

(30)



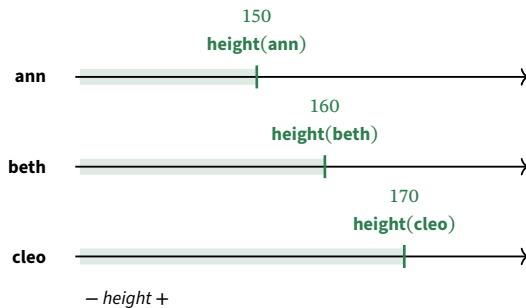
(31)



- Just as determiners denote relations between sets of individuals, we'll take *er* to denote a relation between degree intervals (i.e., $\langle dt, \langle dt, t \rangle \rangle$).

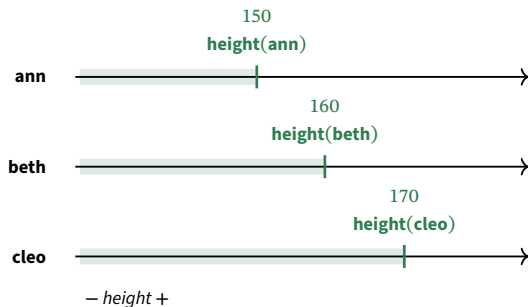
Interlude: ordering intervals

- Suppose the heights of Ann, Beth and Cleo are 150cm, 160cm, and 170cm, respectively.
- Consider the intervals of degrees that each of them is taller or equal to:



Interlude: ordering intervals

- Suppose the heights of Ann, Beth and Cleo are 150cm, 160cm, and 170cm, respectively.
- Consider the intervals of degrees that each of them is taller or equal to:



- These degree intervals are ordered by proper subsethood:

$$(32) \quad \{d \mid \mathbf{height(ann)} \geq d\} \subset \{d \mid \mathbf{height(beth)} \geq d\} \subset \{d \mid \mathbf{height(cleo)} \geq d\}$$

Meaning of the comparatives

- We can then assign the following meaning to the comparative morpheme:

$$(33) \quad \llbracket \text{er} \rrbracket = \lambda D_{\text{than}}. \lambda D_{\text{matrix}}. D_{\text{than}} \subset D_{\text{matrix}}$$

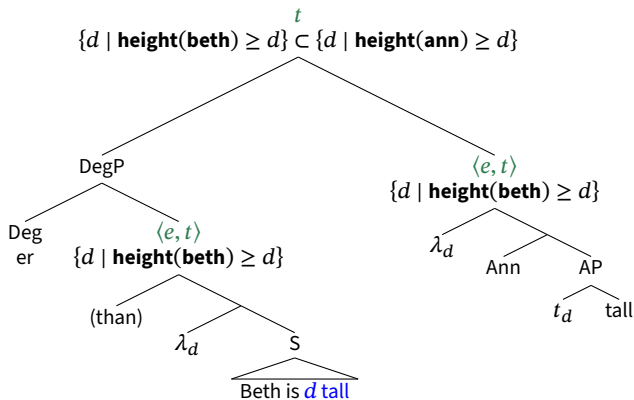
Meaning of the comparatives

- We can then assign the following meaning to the comparative morpheme:

$$(33) \quad \llbracket \text{er} \rrbracket = \lambda D_{\text{than}} \cdot \lambda D_{\text{matrix}} \cdot D_{\text{than}} \subset D_{\text{matrix}}$$

- Interpreting a comparative sentence:

(34)



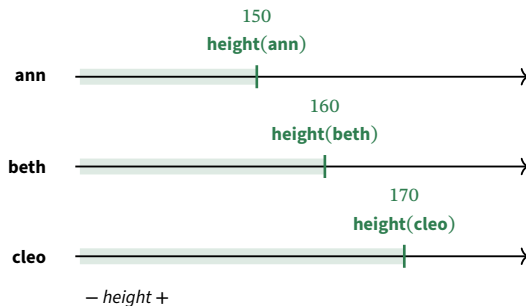
Alternative formulation

- Another approach to meaning of the comparative is as follows:

(Stechow 1984)

$$(35) \quad \llbracket er \rrbracket = \lambda D_{\text{than}} \cdot \lambda D_{\text{matrix}} \cdot \mathbf{max}(D_{\text{than}}) < \mathbf{max}(D_{\text{matrix}})$$

- This meaning involves direct comparison of degrees, but yields the same results.



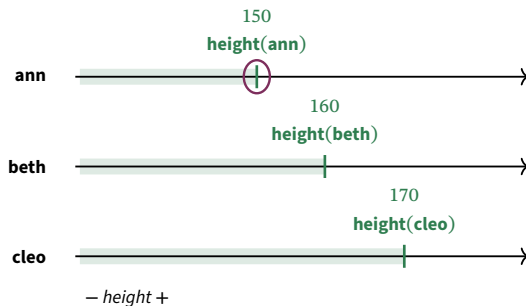
Alternative formulation

- Another approach to meaning of the comparative is as follows:

(Stechow 1984)

$$(35) \quad \llbracket \text{er} \rrbracket = \lambda D_{\text{than}} \cdot \lambda D_{\text{matrix}} \cdot \mathbf{max}(D_{\text{than}}) < \mathbf{max}(D_{\text{matrix}})$$

- This meaning involves direct comparison of degrees, but yields the same results.



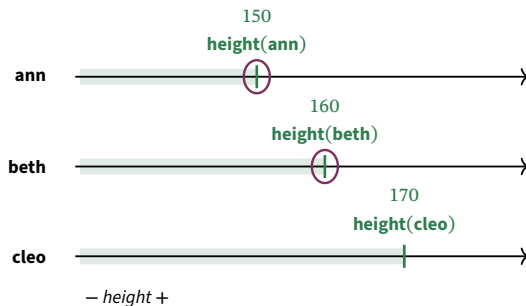
Alternative formulation

- Another approach to meaning of the comparative is as follows:

(Stechow 1984)

$$(35) \quad \llbracket \text{er} \rrbracket = \lambda D_{\text{than}}. \lambda D_{\text{matrix}}. \mathbf{max}(D_{\text{than}}) < \mathbf{max}(D_{\text{matrix}})$$

- This meaning involves direct comparison of degrees, but yields the same results.



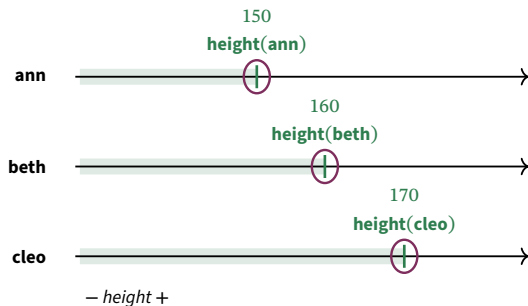
Alternative formulation

- Another approach to meaning of the comparative is as follows:

(Stechow 1984)

$$(35) \quad \llbracket \text{er} \rrbracket = \lambda D_{\text{than}} \cdot \lambda D_{\text{matrix}} \cdot \mathbf{max}(D_{\text{than}}) < \mathbf{max}(D_{\text{matrix}})$$

- This meaning involves direct comparison of degrees, but yields the same results.



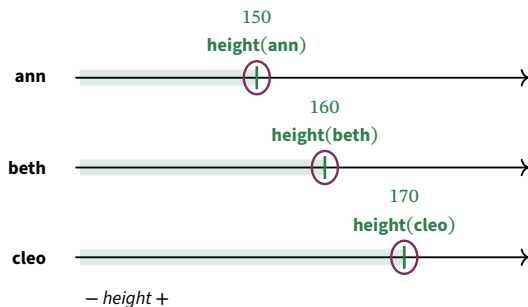
Alternative formulation

- Another approach to meaning of the comparative is as follows:

(Stechow 1984)

$$(35) \quad \llbracket \text{er} \rrbracket = \lambda D_{\text{than}} \cdot \lambda D_{\text{matrix}} \cdot \mathbf{max}(D_{\text{than}}) < \mathbf{max}(D_{\text{matrix}})$$

- This meaning involves direct comparison of degrees, but yields the same results.



$$(36) \quad \mathbf{heigh}(\mathbf{ann}) < \mathbf{height}(\mathbf{beth}) < \mathbf{height}(\mathbf{cleo})$$

Sub-comparatives

- Our analysis can also be easily extended to sub-comparatives:

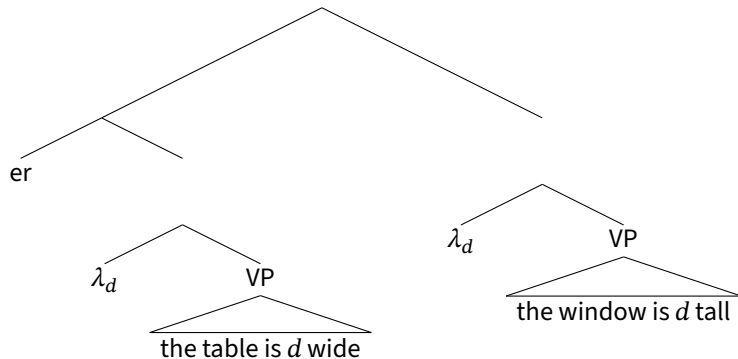
(37) The window is taller than the table is wide.

Sub-comparatives

- Our analysis can also be easily extended to sub-comparatives:

(37) The window is taller than the table is wide.

(38)

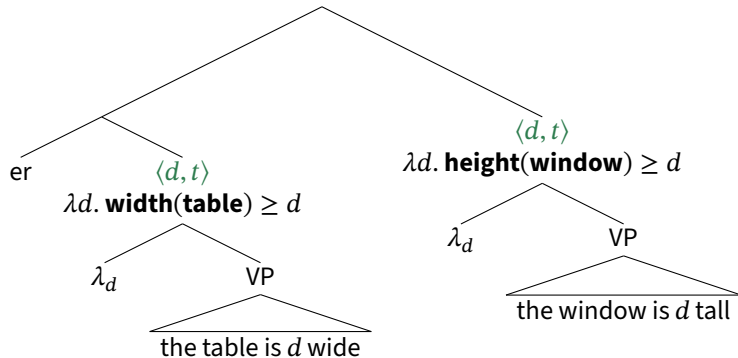


Sub-comparatives

- Our analysis can also be easily extended to sub-comparatives:

(37) The window is taller than the table is wide.

(38)



Part 6

Cross-polar anomalies

Comparatives with antonyms

- Remember that by using *short* we encode a different set of intervals:

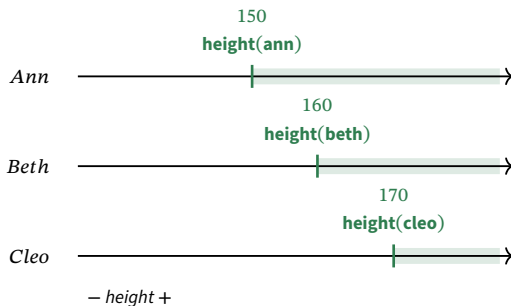
$$(39) \llbracket \text{short} \rrbracket = \lambda d. \lambda x. d \geq \mathbf{height}(x)$$

Comparatives with antonyms

- Remember that by using *short* we encode a different set of intervals:

$$(39) \llbracket \text{short} \rrbracket = \lambda d. \lambda x. d \geq \mathbf{height}(x)$$

- Note that these degrees are still ordered by the subset relation — but the other way around:



$$\{d \mid d \geq \mathbf{height}(\text{cleo})\} \subset \{d \mid d \geq \mathbf{height}(\text{beth})\} \subset \{d \mid d \geq \mathbf{height}(\text{ann})\}$$

Accounting for equivalencies

- Remember we discussed the following are equivalent:
(40) a. Ann is taller than Beth.
b. Beth is shorter than Ann.

Accounting for equivalencies

- Remember we discussed the following are equivalent:

- (40)
- Ann is taller than Beth.
 - Beth is shorter than Ann.

- We account for this:

- (41)
- Ann is taller than Beth.
 - $\{d \mid \text{height}(\text{beth}) \geq d\} \subset \{d \mid \text{height}(\text{ann}) \geq d\}$



Accounting for equivalencies

- Remember we discussed the following are equivalent:

- (40) a. Ann is taller than Beth.
b. Beth is shorter than Ann.

- We account for this:

- (41) a. Ann is taller than Beth.
b. $\{d \mid \text{height}(\text{beth}) \geq d\} \subset \{d \mid \text{height}(\text{ann}) \geq d\}$



- (42) a. Beth is shorter than Ann.
b. $\{d \mid d \geq \text{height}(\text{ann})\} \subset \{d \mid d \geq \text{height}(\text{beth})\}$



Cross-polar anomalies

- It has been observed that the following are not acceptable:
(Bierwisch 1989, Hale 1970, Seuren 1978, Kennedy 2001)
- (43) a. *Ann is shorter than Beth is tall.
b. *Beth is taller than Beth is short.

Cross-polar anomalies

- It has been observed that the following are not acceptable:
(Bierwisch 1989, Hale 1970, Seuren 1978, Kennedy 2001)

- (43) a. *Ann is shorter than Beth is tall.
b. *Beth is taller than Beth is short.

- What does our analysis of antonyms and comparatives has to say about these?

- (44) [[Ann is shorter than Beth is tall]]

Cross-polar anomalies

- It has been observed that the following are not acceptable:
(Bierwisch 1989, Hale 1970, Seuren 1978, Kennedy 2001)

- (43) a. *Ann is shorter than Beth is tall.
b. *Beth is taller than Beth is short.

- What does our analysis of antonyms and comparatives has to say about these?

- (44) $\llbracket \text{Ann is shorter than Beth is tall} \rrbracket$
 $= \llbracket \text{er} \rrbracket (\{d \mid \text{height}(\text{beth}) \geq d\}) (\{d \mid d \geq \text{height}(\text{ann})\})$

Cross-polar anomalies

- It has been observed that the following are not acceptable:
(Bierwisch 1989, Hale 1970, Seuren 1978, Kennedy 2001)

- (43) a. *Ann is shorter than Beth is tall.
b. *Beth is taller than Beth is short.

- What does our analysis of antonyms and comparatives has to say about these?

- (44) $\llbracket \text{Ann is shorter than Beth is tall} \rrbracket$
 $= \llbracket \text{er} \rrbracket (\{d \mid \text{height}(\text{beth}) \geq d\}) (\{d \mid d \geq \text{height}(\text{ann})\})$
 $= \{d \mid \text{height}(\text{beth}) \geq d\} \subset \{d \mid d \geq \text{height}(\text{ann})\}$

Cross-polar anomalies

- It has been observed that the following are not acceptable:
(Bierwisch 1989, Hale 1970, Seuren 1978, Kennedy 2001)

- (43) a. *Ann is shorter than Beth is tall.
b. *Beth is taller than Beth is short.

- What does our analysis of antonyms and comparatives has to say about these?

- (44) $\llbracket \text{Ann is shorter than Beth is tall} \rrbracket$
 $= \llbracket \text{er} \rrbracket (\{d \mid \text{height}(\text{beth}) \geq d\}) (\{d \mid d \geq \text{height}(\text{ann})\})$
 $= \{d \mid \text{height}(\text{beth}) \geq d\} \subset \{d \mid d \geq \text{height}(\text{ann})\}$

$\leadsto$ *this can never be true!*

Adding a presupposition

- To make things concrete, let's add a definedness conditions to er :

(45) $\llbracket er \rrbracket(D)(D')$ is defined only if $D \subseteq D'$ or $D' \subseteq D$.
When defined, $\llbracket er \rrbracket(D)(D')$ is true iff $D \subset D'$.

Cross-polar “nomalies”

- The picture of cross-polar anomalies is, however, more complicated.

Cross-polar “nomalies”

- The picture of cross-polar anomalies is, however, more complicated.
- **Büring 2007**: these sentences are grammatical when the same antonym pair isn't involved.

- (46)
- a. The ladder is shorter than the house is high.
 - b. The rope is shorter than the gap is wide.

Cross-polar “nomalies”

- The picture of cross-polar anomalies is, however, more complicated.
- **Büring 2007**: these sentences are grammatical when the same antonym pair isn't involved.

- (46) a. The ladder is shorter than the house is high.
b. The rope is shorter than the gap is wide.

- Notice that it's essential that the marked antonym is in the matrix clause:

- (47) *The house is higher than the ladder is short.

Cross-polar “nomalies”

- The picture of cross-polar anomalies is, however, more complicated.
- **Büring 2007**: these sentences are grammatical when the same antonym pair isn't involved.

- (46) a. The ladder is shorter than the house is high.
b. The rope is shorter than the gap is wide.

- Notice that it's essential that the marked antonym is in the matrix clause:

- (47) *The house is higher than the ladder is short.

- The general assessment about these data is not that our analysis is incorrect, but rather antonyms are syntactically complex.
- For more details, see Büring 2007 and Heim 2008.

A desired result

- Consider now:

(48) *Ann is taller than Beth isn't.

- What do we have to say about such sentences?

Part 7

Summing things up

Summary

- Today, we finally tackled comparative constructions:
 - the syntax of comparatives:
 - * *than*P is an extraposed complement of *er*
 - * clausal *than*Ps involve covert DegP movement
 - * *er* also covertly moves in the matrix clause

Summary

- Today, we finally tackled comparative constructions:
 - the syntax of comparatives:
 - * *than*P is an extraposed complement of *er*
 - * clausal *than*Ps involve covert DegP movement
 - * *er* also covertly moves in the matrix clause
 - the semantics of comparatives
 - * *er* denotes a relation between intervals
 - * our analysis covers comparatives with either antonym pair
 - * and also why cross-polar anomalies are anomalies to begin with

References I

- Bhatt, Rajesh and Roumyana Pancheva (2004).** “Late Merger of Degree Clauses”. In: *Linguistic Inquiry* 35.1, pp. 1–45.
- Bierwisch, Manfred (1989).** “The Semantics of Gradation”. In: *Dimensional Adjectives*. Ed. by Manfred Bierwisch and Ewald Lang. Berlin: Springer-Verlag, pp. 71–261.
- Bresnan, Joan W. (1973).** “Syntax of the Comparative Clause Construction in English”. In: *Linguistic Inquiry* 4.3, pp. 275–343. URL: <https://www.jstor.org/stable/4177775>.
- Büring, Daniel (2007).** “Cross-Polar Nomalies”. In: *Proceedings of Semantics and Linguistic Theory* 17. Ed. by Tova Friedman and Masayuki Gibson. Ithaca, NY: CLC Publications, pp. 37–52. URL: <https://doi.org/10.3765/salt.v17i0.2957>.
- Chomsky, Noam (1965).** *Aspects of the theory of syntax*. Cambridge, MA: MIT Press.
- **(1977).** “On Wh-movement”. In: *Formal syntax*. Ed. by Peter Culicover, Thomas Wasow, and Adrian Akmajian. New York: Academics Press, pp. 72–132.
- Hale, A. (1970).** “Conditions on English Comparative Clause Pairings”. In: *Readings in English Transformational Grammar*. Ed. by R. A. Jacobs and P. S. Rosenbaum. Waltham, Massachusetts: Ginn, pp. 30–55.
- Heim, Irene (2001).** “Degree Operators and Scope”. In: *Audiatur Vox Sapientiae: A Festschrift for Arnim von Stechow*. Ed. by Caroline Féry and Wolfgang Sternefeld. Berlin: Akademie Verlag, pp. 214–239.
- **(2008).** “Decomposing Antonyms?” In: *Proceedings of Sinn und Bedeutung* 12. Ed. by Atle Grønn. Oslo: ILOS, pp. 212–225.
- Kennedy, Christopher (2001).** “Polar Opposition and the Ontology of “Degrees””. In: *Linguistics and Philosophy* 24.1, pp. 33–70.
- Selkirk, Elisabeth (1970).** “On the Determiner Systems of Noun Phrases and Adjective Phrases”. Unpublished mimeograph, Massachusetts Institute of Technology, Cambridge, Massachusetts.
- Seuren, Pieter A. M. (1978).** “The Structure and Selection of Positive and Negative Gradable Adjectives”. In: *Proceedings of the Fourteenth Annual Meeting of the Chicago Linguistic Society*. Chicago, IL: Chicago Linguistic Society, pp. 336–346.
- Stechow, Arnim von (1984).** “Comparing Semantic Theories of Comparison”. In: *Journal of Semantics* 3.1, pp. 1–77.

Part 8

Disjunction in comparatives

The puzzle

- In our first class, we discussed the following puzzle:
- (49) Ann is taller than either Beth or Cleo are.
 $\rightsquigarrow$ *Ann is taller than Beth & Ann is taller than Cleo*

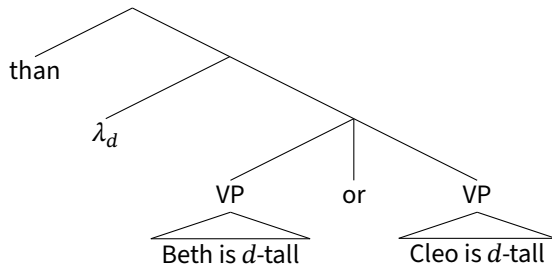
The puzzle

- In our first class, we discussed the following puzzle:
(49) Ann is taller than either Beth or Cleo are.
 $\rightsquigarrow$ *Ann is taller than Beth & Ann is taller than Cleo*
- We are now in a position to account for this fact!

than-clauses with disjunction

- Consider the following *than*-clause:

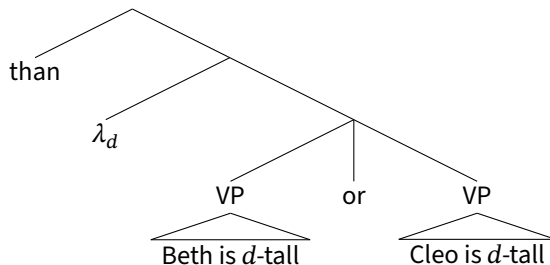
(50)



than-clauses with disjunction

- Consider the following *than*-clause:

(50)



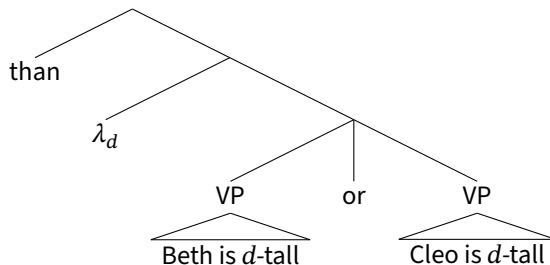
- This interval will denote the **union** of the Beth- and the Cleo-interval:

$$\begin{aligned} (51) \quad & \{d \mid \mathbf{height}(\mathbf{beth}) \geq d \vee \mathbf{height}(\mathbf{cleo}) \geq d\} \\ & = \{d \mid \mathbf{height}(\mathbf{beth}) \geq d\} \cup \{\mathbf{height}(\mathbf{cleo}) \geq d\} \end{aligned}$$

than-clauses with disjunction

- Consider the following *than*-clause:

(50)



- This interval will denote the **union** of the Beth- and the Cleo-interval:

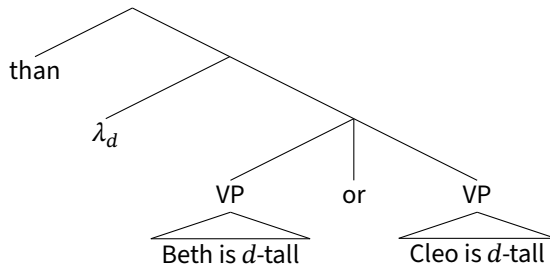
$$\begin{aligned} (51) \quad & \{d \mid \mathbf{height}(\mathbf{beth}) \geq d \vee \mathbf{height}(\mathbf{cleo}) \geq d\} \\ &= \{d \mid \mathbf{height}(\mathbf{beth}) \geq d\} \cup \{\mathbf{height}(\mathbf{cleo}) \geq d\} \end{aligned}$$



than-clauses with disjunction

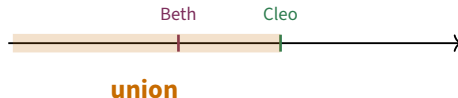
- Consider the following *than*-clause:

(50)



- This interval will denote the **union** of the Beth- and the Cleo-interval:

(51) $\{d \mid \text{height}(\text{beth}) \geq d \vee \text{height}(\text{cleo}) \geq d\}$
 $= \{d \mid \text{height}(\text{beth}) \geq d\} \cup \{\text{height}(\text{cleo}) \geq d\}$



Interpreting the full sentence

- Interpreting the full sentence:

(52) a. Ann is taller than either Beth or Cleo are.

b. $\{d \mid \text{height}(\text{beth}) \geq d\} \cup \{\text{height}(\text{cleo}) \geq d\} \subset \{d \mid \text{height}(\text{ann}) \geq d\}$

